

Project Design Phase

Problem – Solution Fit Template

Date	2 July 2026
Team ID	
Project Name	India's Agricultural Crop Production Analysis
Maximum Marks	2 Marks

Problem – Solution Fit Template:

The **Problem–Solution Fit** for this project focuses on identifying the challenges faced by stakeholders in analyzing India's agricultural crop production data and developing an interactive Tableau dashboard that effectively addresses those challenges. The solution transforms complex agricultural datasets into meaningful visual insights, enabling faster analysis, better decision-making, and improved agricultural planning.

Purpose:

- To simplify the analysis of large-scale agricultural crop production data through interactive data visualization.
- To help government agencies, researchers, policymakers, and farmers make informed decisions using accurate and easy-to-understand dashboards.
- To identify trends in crop production, cultivated area, and yield across different states and years.
- To provide a centralized platform for comparing agricultural performance using dynamic charts, maps, and KPIs.
- To improve data accessibility and support evidence-based agricultural planning and policy formulation.

Template:

Problem–Solution fit canvas 2.0			Purpose / Vision: Enable data-driven agricultural planning by transforming historical crop production data into interactive visual insights for better decision-making.			
Define CS, fit into CC	1. CUSTOMER SEGMENT(S) Who is your customer? - Ministry of Agriculture & State Agriculture Departments - Policy Makers & Government Officials - Agricultural Researchers & Academicians - Farmers & Farmer Producer Organizations (FPOs) - NGOs & Development Organizations - Students & Data Analysts	CS	6. CUSTOMER CONSTRAINTS What constraints prevent your customers from taking action or limit their choices of solutions? i.e. spending power, budget, no cash, network connection, available devices. - Large and complex agricultural datasets - Scattered data across years and states - Time-consuming manual analysis - Limited visualization and reporting tools - Lack of real-time, interactive insights - Limited technical expertise in data analysis	CC	5. AVAILABLE SOLUTIONS Which solutions are available to the customers when they face the problem or need to get the job done? What have they tried in the past? What pros & cons do these solutions have? i.e. pen and paper is an alternative to digital notetaking. - Government agricultural reports (PDF) - Excel spreadsheets and pivot tables - Static charts and basic reports - Manual calculations and summary tables - Other BI tools with limited interactivity	AS
	2. JOBS-TO-BE-DONE / PROBLEMS Which jobs-to-be-done (or problems) do you address for your customers? There could be more than one; explore different sides. - Analyze crop production trends over years - Compare agricultural production across states - Analyze cultivated area and productivity - Identify top performing crops and states - Support policy making and planning - Monitor yield trends and performance	J&P	9. PROBLEM ROOT CAUSE What is the real reason that this problem exists? What is the back story behind the need to do this job? i.e. customers have to do it because of the change in regulations. - Agricultural data is available in large volumes but scattered across sources - Data is not in an easy-to-interpret format - Lack of interactive dashboards for comparison - Manual report generation is slow and inefficient - Decision-makers lack timely, actionable insights	RC	7. BEHAVIOUR What does your customer do to address the problem and get the job done? i.e. directly related: find the right solar panel installer, calculate usage and benefits; indirectly associated: customers spend free time on volunteering work (i.e. Greenpeace) - Download agricultural datasets from government sources - Use Excel for data cleaning and analysis - Read annual agricultural reports - Compare production and area manually - Search for crop and state performance information - Share reports with stakeholders	BE
Identify strong TR & EM	3. TRIGGERS What triggers customers to act? i.e. seeing their neighbour installing solar panels, reading about a more efficient solution in the news. - Need for data-driven agricultural planning - Government policy reviews and planning cycles - Rising food demand and population growth - Climate change impact on agriculture - Requirement for academic research and analysis	TR	10. YOUR SOLUTION If you are working on an existing business, write down your current solution first, fill in the canvas, and check how much it fits reality. If you are working on a new business proposition, then keep it blank until you fill in the canvas and come up with a solution that fits within customer limitations, solves a problem and matches customer behaviour. An interactive Tableau dashboard that provides: - KPI cards – Total Production, Total Area, Average Yield - State-wise production distribution (Map) - Crop-wise cultivated area comparison - Year-wise production trend analysis - Yield analysis and performance metrics - Interactive filters – Year, State, Season, Crop - Drill-downs and tooltips for detailed insights - Easy sharing and decision-making support	SL	8. CHANNELS of BEHAVIOUR 8.1 ONLINE What kind of actions do customers take online? Extract online channels from #7 - Government Open Data Portal - Tableau Public Dashboard - GitHub Repository - Research websites and publications - Kaggle / Data platforms	CH
	4. EMOTIONS: BEFORE / AFTER How do customers feel when they face a problem or a job and afterwards? i.e. lost, insecure > confident, in control – use it in your communication strategy & design. Before: Confused, Overwhelmed, Uncertain, Time-consuming analysis After: Clear insights, Confident, Informed decisions, Satisfied, Saves time and effort	EM			8.2 OFFLINE What kind of actions do customers take offline? Extract offline channels from #7 and use them for customer development. - Agriculture Offices and Departments - Universities and Research Institutions - Farmer Meetings and Workshops - Government Committees and Meetings - Printed reports and presentations	